

The form of Viscid Burgers' Equation from 1D to 3D

_____ 1D:

$$u_t + uu_x = \nu u_{xx}$$

where

$$u = u^x(x, t)$$

_____ 2D:

$$u_t^x = -u^x u_x^x - u^y u_y^x + \nu(u_{xx}^x + u_{yy}^x) \quad (1)$$

$$u_t^y = -u^x u_x^y - u^y u_y^y + \nu(u_{xx}^y + u_{yy}^y) \quad (2)$$

where

$$u^x = u^x(x, y, t)$$

$$u^y = u^y(x, y, t)$$

_____ 3D:

$$u_t^x = -u^x u_x^x - u^y u_y^x - u^z u_z^x + \nu(u_{xx}^x + u_{yy}^x + u_{zz}^x) \quad (3)$$

$$u_t^y = -u^x u_x^y - u^y u_y^y - u^z u_z^y + \nu(u_{xx}^y + u_{yy}^y + u_{zz}^y) \quad (4)$$

$$u_t^z = -u^x u_x^z - u^y u_y^z - u^z u_z^z + \nu(u_{xx}^z + u_{yy}^z + u_{zz}^z) \quad (5)$$

where

$$u^x = u^x(x, y, z, t)$$

$$u^y = u^y(x, y, z, t)$$

$$u^z = u^z(x, y, z, t)$$